

Technical Logic Validation — 7-Point Checklist + Log

Day 4 handout. Each artifact can be fine in isolation yet inconsistent together. Technical logic validation = **walk every cross-artifact reference** and verify they fit. Most quads fail 2–3 checks on first pass — that's expected; that's why this day exists.

The failure this catches

- The PRD's NFR says "p95 < 500ms" but the TMD's sequence diagram totals 800ms.
 - The TCD says "modular monolith" but the TMD shows components that need to deploy independently.
 - The data model has no foreign key for a relationship the API requires.
 - The outcome map's "leading indicator" isn't measurable with the observability NFR.
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The 7-point validation checklist

#	Check	Where to look
1	Outcome map → metrics → SLOs aligned	DP-light Section 1, PRD-light Section 7, TCD-light Section 4
2	Data model supports the API contract	TMD-light Section 1 ↔ Section 3
3	API contract supports the AC	TMD-light Section 3 ↔ PRD-light Section 6
4	SLO budget fits the sequence diagram	TCD-light Section 4 ↔ TMD-light Section 4
5	NFRs don't contradict the architecture stance	PRD-light Section 7 ↔ TCD-light Section 1
6	Trade-offs don't contradict each other	TCD-light Section 5
7	Stakeholder sign-off captures all constraints	TCD-light Section 6

For each check: pull **both sides**, compare, and if they disagree — decide to tighten the artifact (usually the answer), adjust the SLO/NFR only if evidence demands, or mark it a known gap to call out in the presentation.

The validation log

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| # | Check | Status | Gap / fix |
|---|-----|-----|-----|
| 1 | Outcome → metrics → SLOs | Pass | None |
| 2 | Data model ↔ API | Partial | Missing FK on
User → Order; added |
| 3 | API ↔ AC | Pass | None |
| 4 | SLO budget ↔ sequence | Fail → Fixed |
Sequence totaled 800ms; SLO was 500ms; tightened SLO
to p95 < 1000ms with defense |
| 5 | NFRs ↔ architecture | Pass | None |
| 6 | Trade-offs internal consistency | Pass | None |
|
| 7 | Stakeholders ↔ constraints | Partial |
Compliance not on sign-off list; added |
```

(The example rows above show the log's shape — replace them with your own findings.)

The bar to clear

Don't mark a check "Pass" without walking the references — be ready to **show the section reference** for every pass. A quad that genuinely passes all 7 has a coherent artifact set.